Learning Minecraft Parkour That Transfers to Unseen Courses

1. Introduction

Parkour in Minecraft is a sequence of precise jumps between small blocks over a void. A player must judge distance, height and direction, build up speed, and press jump on the right game tick; one mistake ends the run. We ask whether a reinforcement-learning agent can learn this skill well enough to complete **courses it has never seen**, and whether what it learns in simulation holds in the real game.

Our contributions:

1. A GPU simulator matching prismarine-physics, a Mineflayer bot's movement engine, to 1.4e-14 blocks (section 3).
2. A study of how forgiving 243 jump types are, defining six levels and a curriculum (sections 4 and 5).
3. A PPO agent that completes held-out courses, with ablations of our first reward ideas (distance to the goal, number of jumps, height), the curriculum and the route input (sections 5 and 7).
4. A Mineflayer bridge running the network in real Minecraft, checked tick by tick against the simulator (sections 6 and 7.6).
5. Neo jumps around a wall's end, which exploration never found: learned from search demonstrations and DAgger (section 7.7), and planned at play time with the exact simulator, narrowly meeting our 85% single-neo goal (section 7.8).
6. Routing without being told the platforms: a greedy planner, and a learned router that beats it (section 7.9).
7. One network for every skill, matching the specialists, and a learned skill gate (section 7.10).

2. Related work

- **Reinforcement learning in Minecraft** has mostly targeted survival and crafting from human data (MineRL, Guss et al., 2019; VPT, Baker et al., 2022). We isolate movement and replace human data with a faithful simulator.
- **Learned parkour and locomotion.** Varied terrain makes agile locomotion emerge from simple rewards (Heess et al., 2017), and robot parkour trained in parallel simulation transfers to real robots (Zhuang et al., 2023; Cheng et al., 2024). We follow that recipe with the bot's own physics engine reproduced exactly, so the reality gap is measurable: rounding error, except next to wall corners (section 7.8).
- **Massively parallel PPO.** Rudin et al. (2021) trained legged robots in minutes on thousands of GPU environments. We use their KL-adaptive learning rate with PPO (Schulman et al., 2017), GAE (Schulman et al., 2016) and dual clipping (Ye et al., 2020).
- **Generalisation and curricula.** Procgen (Cobbe et al., 2020) showed that agents trained on few levels overfit; we train on endlessly generated courses and test on disjoint seed families. Curricula (Bengio et al., 2009) motivate our level schedule, and potential-based shaping (Ng et al., 1999) the progress reward.
- **Search and hierarchy.** Our neo searches use the cross-entropy method (Rubinstein, 1999); DAgger (Ross et al., 2011) queries them in the learner's states. The learned router and the gate make one choice per landing: an option in a semi-MDP (Sutton et al., 1999). The one network distils the specialists (Rusu et al., 2016), and the gate is a mixture of frozen experts (Jacobs et al., 1991).

3. A faithful simulator

Why it can be faithful. A Mineflayer bot computes its own movement on the client with prismarine-physics and sends the resulting position to the server. The server re-applies the move with the game's own collision order; in 40 recorded runs it sent the bot back (a correction) only when the two results differed by more than 0.25 blocks, which happened only next to wall corners (section 7.8). Otherwise the bot's movement in real

Minecraft is prismarine-physics. The bot never applies a 1.21 server's extra sprint modifier, so it sprints like the vanilla client and the simulator.

The port. parkour/physics.py re-implements prismarine-physics 1.11.1 (PrismarineJS), batched in PyTorch, in float64, keeping the operand order of the JavaScript source: ground and air acceleration, sprinting, the sprint-jump boost, drag, gravity, the jump cooldown, axis-by-axis collision against block boxes, and step-up. Levels 0 to 5 contain only air and full cubes. Ice, block shapes (slabs, stairs, fence posts and more), slime and ladders are opt-in, each exact against prismarine-physics (results/physics_parity_*.json); water is not modelled.

Validation. parkour/parity.py runs prismarine-physics through Node, with the 1e-7 collision tolerance that the simulator and the bot share (vanilla's rule), on random block fields (pillars, steps, ceilings, gaps, edges) with random inputs, and compares every tick: over 99,000 ticks on the CPU and the GPU (results/physics_parity.json) the largest position difference was 1.4e-14 blocks, and every on-ground, collision and cooldown flag matched.

Speed. The environment step never reads values back from the GPU. With 16,384 parallel environments, training on one RTX 5080 runs at about 170,000 to 220,000 game ticks per second over whole runs, evaluations included (main-s0, main-g995 and obstacles-g995 in results/runs/).

4. Courses and difficulty

A course is a chain of stone platforms in the void. Each jump has a gap (0 to 5 air blocks), a height change (-3 to +1), a sideways offset (0 to 2) and a landing size (1x1, 2x2 or 3x3).

Feasibility study. For each of 243 jump types we simulate 4,032 scripted strategies from a standing start on a 1x1 block (start position, aim point, jump tick, sprint or walk, forward release). The fraction that lands is the jump's *forgiveness*. Scripted strategies land 194 types; flat 1-, 3-, 4- and 5-block gaps have forgiveness 0.47, 0.08, 0.01 and 0. Six forgiveness bands define levels 0 to 5.

Generation. A level-L course draws half its jumps from level L and half from easier levels, turns with probability 0.25, and keeps one air block between non-consecutive platforms and each flight path clear of earlier platforms. Training courses have 5 to 12 jumps, test courses 10 (fewer if no jump fits).

Held-out sets. Each set has its own seed family; training uses family 0, minus any course equal to a held-out one. Standard and obstacle courses (a head-hitter beam or decoy pillar on half the jumps): test families 1 and 4 (200 per level), validation 2 and 5 (100 per level; larger draws in sections 7.7 to 7.9). Single-neo courses (one neo among ten jumps): test 41, validation 40 (320 each, section 7.8). Level 5 with and without obstacles: validation only, 46 and 47 (2,000 each, section 7.5). Ice on 30% of platforms: test 16, validation 17 (section 7.10).

5. MDP and learning

State. The course; the player's float64 position and velocity, yaw, on-ground and collision flags and jump cooldown; the progress index *k* (the furthest platform stood on); tick and stall counters; the previous action.

Observation (255 numbers). In the player's frame: velocity, on-ground, jump cooldown, horizontal collision (6); previous action (5); platforms *k* to *k*+3, each: valid and goal flags, four top-face corners, top height (44); floor height and ceiling flag at 98 points from 1 block behind to 5.5 ahead, 1.5 to each side (196); goal offset (3); fraction of platforms reached (1). Later platforms (except the goal), obstacles outside the scan and the counters are hidden: a POMDP, played by a memoryless policy.

Actions (252). Factored: move (back, none, forward) x strafe (left, none, right) x jump x sprint x yaw step (-20, -6, -1.5, 0, 1.5, 6, 20 degrees, applied before the tick's movement).

Dynamics. Deterministic: each 50 ms step is one tick of the float64 port of prismarine-physics. Randomness: the course drawn at reset and, in training, the sampled actions.

Initial state. A level from the curriculum (below), a training course of that level, the player at rest on platform 0's centre, facing within 60 degrees of platform 1 (fixed per course), *k* and the counters 0, no previous input.

Discount. 0.995 (final policy, section 7.3; default 0.99).

Reward and episode ends (RewardConfig defaults, as in the final policy; our first ideas in brackets):

Term (idea)	Reward	Episode
progress (distance to the goal)	decrease of the path left to the goal (feet to the next platform's top centre, then centre to centre), weight 1	continues
landing (number of jumps)	+1 for each platform *k* advances (skipped ones count)	continues
goal	+10 on standing on the last platform	terminal
fall (height)	-5 when the feet are more than 1 block below the lowest top of platforms *k*-1 to *k*+2, or the player stands on an obstacle	terminal
stall	0 after 200 ticks without a new platform (-5 in section 7.7)	truncated, value bootstrapped (terminal if penalised)
time	-0.01 per tick	continues

A 3,000-tick truncation never binds: the stall rule caps episodes at 2,400 ticks.

Algorithm. PPO with GAE (lambda 0.95), separate actor and critic MLPs (512-256-128, ELU), observation standardisation, 16,384 environments, 32-tick rollouts, 4 epochs of 8 minibatches, clip 0.2, entropy 0.005. The level-5 fine-tunes, DAgger round 4 and the runs of section 7.10 used 8,192 environments, 64-tick rollouts and 32 minibatches (the gate 8). A tabular method cannot generalise across course geometries, so function approximation is required.

Curriculum. Levels up to a frontier are sampled with weight 1, the next level with weight 3; the frontier advances when the next level's success rate reaches 60% over 2,000 episodes.

Stability. Our first runs made three catastrophic updates (approximate KL of 0.3, 1.3 and once infinity; training success fell from about 75-80% to 15-17% within two iterations). We added a log-ratio clamp, dual clipping, a per-minibatch KL stop at 0.05 and a KL-adaptive learning rate (target 0.01). No catastrophic update recurred, though training success still dipped at times.

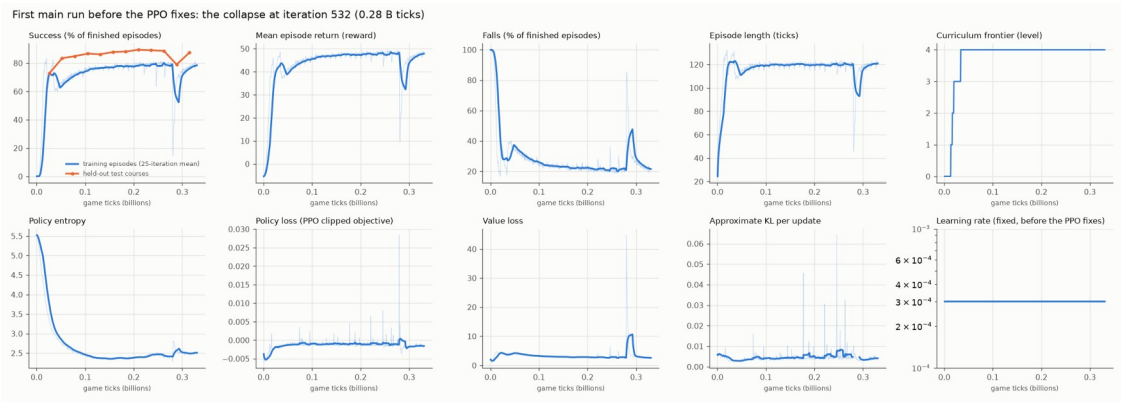


Figure 1: a run before the safeguards (results/runs/_v1_unstable): its catastrophic update (infinite KL, not plotted) shows as a sudden drop in training success.

6. From simulator to Minecraft

The trained actor, exported as JSON, runs in JavaScript inside the bot's physics tick (bridge/policy.js, bridge/parkour_bot.js): after each tick the bot builds the observation from the game, picks the argmax action and sets yaw and controls for the next tick, in the simulator's step order. A local Minecraft 1.21.4 void-world server (parkour/server.py) hosts the runs; the bot builds each course with /fill, teleports onto the start block (the reset) and uses only movement controls during an episode. The same JavaScript on prismarine-physics without a server is the *offline bot*.

The *replay check* re-simulates the recorded actions from the recorded start state and compares positions tick by tick. The *policy check* plays the course in the simulator from the same start, after 10 idle ticks on the start block, and compares outcomes.

7. Results

Every number comes from a file under results/, runs/ or configs/; results/README.md gives an overview. Unless stated otherwise, numbers are simulator results on the held-out test sets (section 4) with the deterministic policy. Checkpoints and settings were chosen on validation courses only. Earlier runs also logged test curves (figure 2) without choosing on them; after 28 Oct each setting's test was played once.

7.1 Main run

With section 5's settings and discount 0.99, the main run trained for 3,000 iterations (1.57 billion game ticks, 2.6 hours on one RTX 5080). Its last checkpoint completes **95.5%** of the 1,200 test courses (level 5: 80%). Levels 0 to 4 pass 90% within 0.08 billion ticks; level 5 reaches 78% by 0.55 billion and then stays between 64 and 92%, apart from one short dip at 1.0 billion (figure 2). Section 7.3 repeats it with discount 0.995.

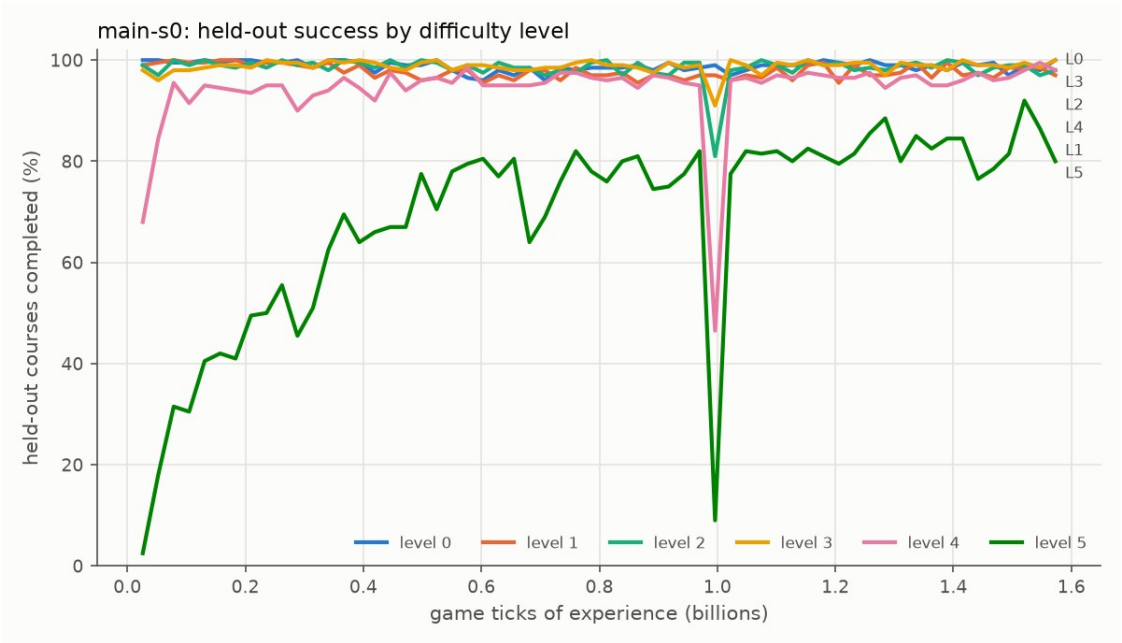


Figure 2: held-out test success per difficulty level during the main run (main-s0), every 50 iterations.

7.2 Reward, curriculum and route ablation (3 seeds per arm, 0.52 B ticks each)

Arm	Final success, mean (min-max)	Level 5, mean (min-max)	Ticks to 80%
full	93.3% (91.9-94.2)	72.0% (69.0-76.5)	0.05 B
no distance shaping (sparse)	93.6% (91.5-95.1)	74.8% (70.0-79.5)	0.14 B
no landing bonus (nolanding)	91.8% (90.0-94.2)	59.5% (47.0-74.5)	0.05 B
no fall penalty (nofall)	90.9% (90.5-91.2)	55.2% (54.0-57.0)	0.05 B
no curriculum (uniform)	94.0% (93.0-95.0)	72.0% (67.0-76.0)	0.08 B
no route: platform slots hidden (noroute)	82.4% (80.6-83.6)	43.0% (39.0-47.5)	0.38 B

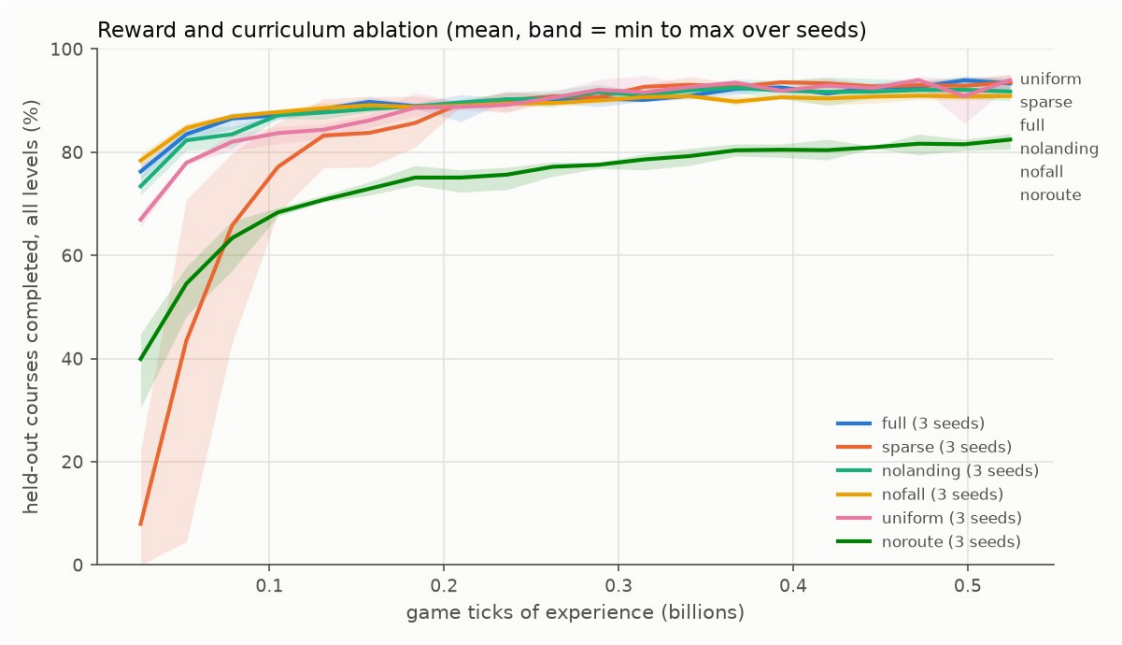


Figure 3: held-out success during training for each ablation arm, 3 seeds per arm.

- Of the reward terms, the **fall penalty** (the height idea) matters most: without it the final success and the level-5 success fall below every seed of the full arm.
- **Distance shaping** does not change the final success but makes learning about three times faster.

- Without the **landing bonus** level-5 success is lower on average, but the seeds overlap, so three seeds show no reliable effect.
- The **curriculum** does not raise the final score at this budget; uniform sampling learns slightly more slowly (0.08 against 0.05 B ticks to 80%) and ends equal.
- Hiding the **route** costs most: the agent needs about seven times as many ticks to reach 80% and ends at 82.4%. Its scan sees 5.5 blocks ahead and 1.5 to each side, so after a turn the next platform is often outside it.

7.3 Hyperparameters and the final policy

One setting at a time around the defaults, seed 0, 0.52 B game ticks each, chosen on validation, test reported (configs/sweep_hparams.json). The defaults score 91.8% on validation (2 seeds) and 93.3% on test (3 seeds, level 5: 72.0%).

Change	Validation	Test	Level 5
discount 0.98	89.2	90.4	58.0
discount 0.995 (3 seeds)	**95.4** (95.2, 95.8, 95.2)	**95.2** (95.0, 94.9, 95.6)	**76.3**
entropy 0.001	94.3	94.1	70.5
entropy 0.02	91.0	91.5	64.0
KL target 0.005	89.7	90.8	52.5
KL target 0.02	92.2	91.8	62.0
network 256-128	92.5	91.3	56.0
network 1024-512-256	94.5	95.4	83.5
4,096 environments (4,000 smaller updates)	91.5	92.0	60.0

Only a longer horizon passed our bar of 3 points over the defaults on validation, in all three seeds (+3.6 validation, +1.8 test, +4.3 level 5). A 10-jump course takes about 120 to 200 ticks; with discount 0.99 the goal bonus is worth 30 to 13% of its value at the start, with 0.995 55 to 37%, so the far goal pulls harder. A shorter horizon (0.98) and a tighter KL target hurt most. The large network is the best single run on level 5 (83.5%), but it is one seed and below the bar on validation.

We then trained discount 0.995 at full length: 3,000 iterations as in the main run (main-g995), then the obstacle fine-tune of section 7.5 (obstacles-g995). Its validation pick (iteration 699) beat the first final policy (obstacles-ft, the same fine-tune of main-s0) there, 97.6% against 96.6%, and became the final policy (results/final_policy.pt). On the test sets it completes 98.4% of standard courses (level 5: 93.5%) against 97.1% (90.5%), and 95.2% of obstacle courses (level 5: 81.0%) against 93.6% (75.5%).

7.4 Jump types and course styles

From a standing start on a 1x1 block at 16 start angles, the final policy makes every jump type of levels 0 to 4 every time and the level-5 types 99.1% of the time. Of the 49 types that none of the 4,032 scripted strategies lands, it makes 3 at least once, including a diagonal 4-block gap onto a 3x3 platform from 15 of the 16 angles (figure 4).

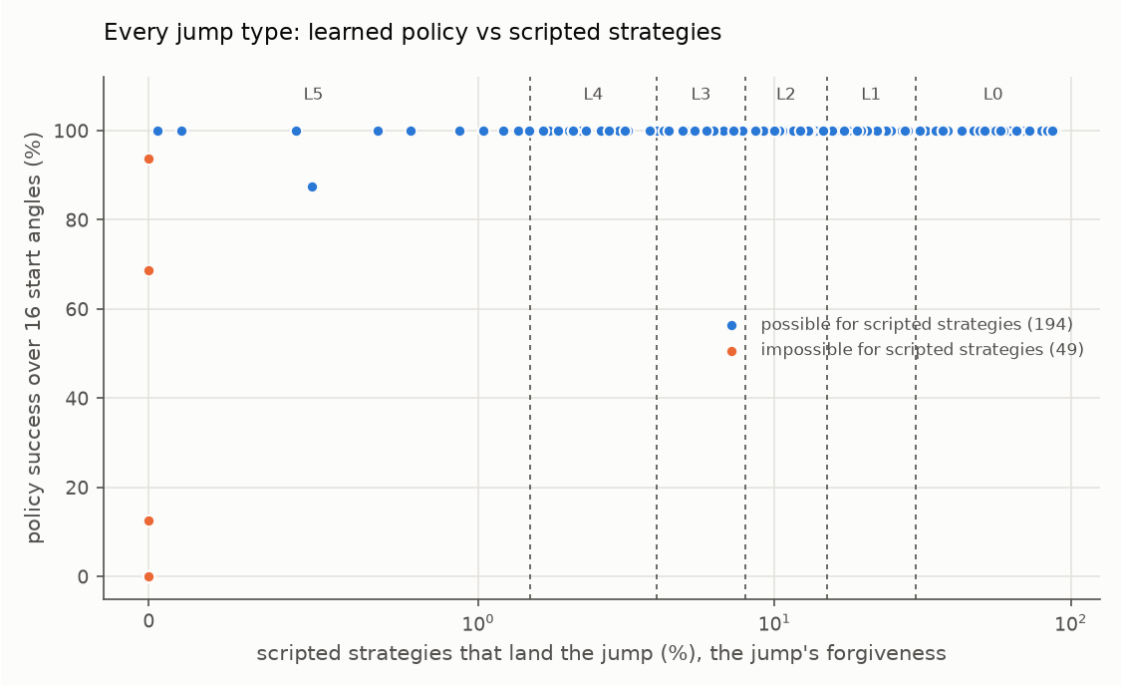


Figure 4: each dot is one jump type: the final policy's success over 16 start angles against the jump's forgiveness (section 4); dashed lines separate the levels.

On course styles outside training (100 new courses per level and style) it completes 93.0% of courses made only of single blocks, 94.2% with a turn at every jump, 96.5% of 15-jump courses, 96.7% from any start direction and 98.3% of straight lines (standard

courses: 98.3%). The weakest cases are level-5 courses of single blocks (75%) or with a turn at every jump (80%).

7.5 Obstacles

Before obstacle training, the main run's policy completed 46.8% of new courses with head-hitter beams and decoy pillars on half the jumps (stress test, 100 per level), against 96.7% of standard ones. The obstacle fine-tune trains for 1,000 iterations with an obstacle on 30% of jumps; after it, the same run completes 95.3% of such courses, and the final policy (the same fine-tune of main-g995, section 7.3) 96.3%. On the obstacle test set, level 5 (81.0%) is its weakest part.

Over three seeds, fine-tuning the final policy with obstacles on half the jumps and level 5 weighted 3 to 1 gains 2.67 points on family 46 (2,000 level-5 validation courses with obstacles, created after the first run of this recipe and of the obstacles-only one had been tested; every seed gains) and lifts the obstacle test set's level 5 from 81.0% to 83.8%, with standard courses unchanged. Each change alone stays within half a point on family 46 but rises about as much on test (82.8% and 84.2%), so the test does not confirm that both are needed; with no fine-tune that has neither change, the test's rise may also come from the extra training itself. We report it but keep the final policy, which the planners, the router and the bots use.

7.6 The bot, offline and in real Minecraft

The offline bot and the bot in real Minecraft (section 6), the latter in real time, each played the final policy on 60 held-out standard courses, 60 held-out obstacle courses and 4 hand-designed courses. Both completed 59, 57 and 3 of them: exactly the courses the simulator completed, with the same tick counts. The failed hand-designed course is built around neo jumps (section 7.7), which this policy never trained on; it stalls at the first neo. The server never corrected the bot's position (0 corrections in 124 episodes), and the replay check reproduced every trajectory to within 7.1e-15 blocks offline and 2.1e-14 blocks in the game. On these courses the gap between simulation and the game is rounding error; at a wall's corner the game's collision order can widen it (section 7.8).

7.7 Neo jumps: refusal, demonstrations and DAgger

What is possible. A neo jumps around the end of a wall: the player takes off beside it, swings around its end in the air and lands on the far side. A cross-entropy search (Rubinstein, 1999) over tick-by-tick actions (parkour/cem.py, configs/jump_cem*.json) landed 16 of 18 single-neo layouts, yet only 1 in 2,000 to 16,000 sampled sequences did: possible, but extremely tight. For double and triple neos (a wall reaching one or two blocks past the platforms' edge) it found no landing in 64 layouts, from rest or a run-up; we treat them as impossible, without proof.

Training. Levels 6 to 8 add single neos, run-up 4-block jumps and wide diagonal jumps. Fine-tuned on levels 0 to 8 for about 1,700 iterations, the discount-0.99 policy obstacles-ft (section 7.3) stayed at about 70% of the level-6 test courses (70.0% before) and rose from 7% to 30% on level 7, then stayed flat; both come from courses without a neo.

Diagnosis: the agent refuses. On each jump type alone, from 16 start angles, the fine-tuned policy makes none of the 13 single-neo types that scripted strategies land: on 11 it stands still until the 200-tick stall limit every time; 95% of its failures are refusals. Under our reward this is rational. A stall truncates the episode with a bootstrapped value, so standing still costs only the time penalty, while a jump that rarely lands mostly earns the fall penalty (-5). The fall penalty that helped most on levels 0 to 5 (section 7.2) here teaches refusal.

A stall penalty does not help. With the optional stall penalty (section 5) a stall ends the episode with -5, but with discount 0.99 a penalty 200 ticks away counts only about a seventh. After 1,000 iterations with it (and entropy 0.01) (neo-stall), the policy still makes none of the 13 types. Exploration almost never finds the curved run a neo needs, so we gave the agent searched landings as demonstrations.

Demonstrations. A second cross-entropy search, over the bot's own 252 actions (parkour/neo_demos.py), found 50,706 landing sequences from standing starts on the 16 landable layouts, wall on either side. neo-bc adds a behaviour-cloning loss on 12,800 of them to PPO; it makes all 13 single-neo types from a standing start, 54.8% of attempts on average.

DAgger. On a course the bot reaches a neo moving, not from rest. DAgger (Ross et al., 2011) labels the states the policy itself visits: the policy plays training courses with neos, and at each neo takeoff it reaches, the search looks for a landing from its exact state. Each found sequence gives imitation pairs (observation, action); each round adds its pairs to the earlier ones and trains for 500 iterations. Two fixes were needed.

- *Label noise.* The first attempt collapsed (from 86.5% to 5% on levels 0 to 8): the search filled ticks where the action does not matter with random actions. It now starts from the policy's own action probabilities and puts the policy's choice back wherever the neo still lands. The fixed rounds also cut the imitation weight from 1 to 0.3, a confound, though neo-bc did not collapse at 1.
- *Neo detection.* Round 1's detector also counted one-block-up jumps (only 1,090 of its 2,526 takeoffs were neos; results/neo/detector_check.json, which counts the collapsed attempt as round 1); from round 2 on, only wall blocks count.

Results on the 210 test courses of levels 6 to 8 with a neo (337 neo jumps, simulator):

Neo policy	Courses completed	Neo jumps passed / reached
stall penalty only (neo-stall)	0/210 (0.0%)	0/202 (0.0%)
+ imitation (neo-bc)	49/210 (23.3%)	82/227 (36.1%)
+ DAgger round 1	62/210 (29.5%)	99/228 (43.4%)
+ DAgger round 2, best checkpoint	63/210 (30.0%)	112/239 (46.9%)
+ DAgger round 2, last checkpoint	74/210 (35.2%)	115/239 (48.1%)
+ DAgger round 3, chosen on the validation neo courses	61/210 (29.0%)	97/224 (43.3%)

DAgger lifts imitation's 23.3% to about 30%, where the rounds level off. The neo policy, round 3's checkpoint chosen on the 218 validation neo courses (75 against 72 for round 2's best), completes 61/210 (29.0%). Round 2's last checkpoint scores higher on test, but checkpoints of one run differ more (63 against 74) than rounds do. A fourth round trailed round 3 on levels 0 to 8 on validation (85.2% against 85.7%) but led on single-neo validation (146 against 133 of 320); section 7.8 uses that snapshot.

Each extra neo is one more tight jump: the neo policy completes 56 of the 129 test courses with one neo, 3 of 47 with two separate neos, 2 of 25 with three and 0 of 9 with more. On 31 neo courses (the first ten test neo courses of each of levels 6 to 8, and a hand-designed one) it completes the same 11 offline, in real Minecraft and in the simulator, with 0 server corrections. Our generator puts every neo after a one-block-wide takeoff, though the search also lands some from 2x2 takeoffs (results/neo/wide_takeoff_search.json). Section 7.10 distils the neo policy, with the other specialists, into one network that also makes neos.

7.8 Planning the hard jumps at play time

The DAgger search (section 7.7) also works at play time as a neo planner (parkour/neo_assist.py). When the bot first stands on a planned takeoff, the search runs from its exact state with the exact simulator, and the bot plays the found sequence until it lands; if nothing lands, it brakes and searches again. The bots check each plan with prismarine-physics. Planning every jump, the final policy, which never trained on neos, completes 147 of the 218 validation neo courses (67.4%; 145 of them taking every neo; alone, 0). Also planning the approach to each takeoff gives 157, but 36 of those skip a neo's takeoff. Its test is on hold.

The single-neo goal, 85% of courses with exactly one neo, came with rules fixed before any result: new single-neo sets (section 4), a strict count (the bot must stand on the neo's takeoff, then land on its target), planner settings tried in a fixed order on validation (each kept only if it added at least 4 courses) and one test run. On validation the neo planner made 271 (84.7%) with the final policy, 267 with the neo policy; up to three search restarts lifted the former to 286 (89.4%), past the 89% (85% plus two standard errors) at which the trials stopped. Played once, this setting completes 276 of 320 test courses (86.25%), against 137 (42.8%) for the best policy alone. The goal is met, narrowly: the search seed alone gives a 320-course count a standard deviation of about 7 to 11 courses (three settings), so 86.25% is one draw and its 4-course margin lies within that noise.

Courses (simulator: strict count; bots: completed)	Final policy + neo planner	Best policy alone (DAgger round 4)
320 validation (family 40), simulator	286 (89.4%)	146 (45.6%), best of 10 snapshots
320 test (family 41), simulator, played once	276 (86.25%)	137 (42.8%)
first 64 test courses, offline bot and real Minecraft	58/64 and 58/64	26/64 and 26/64
the same, with both fixes below	55/64 and 55/64	-

The bots. Before any fix, the offline bot and real Minecraft each completed 58 of the first 64 test courses, as did the policy check (section 6); the simulator test, which starts without the 10 idle ticks, completes 57 of them (56 strict). Two changes followed, prompted by these runs and kept on validation. With the first-jump fix, which also plans from the start state, the server corrected the bot twice and one course failed (54/64; 55 offline). Both came at a neo wall's corner: the game resolves horizontal movement along its larger component first, prismarine-physics along x then z, and the server sends the bot back when the two end over 0.25 blocks apart and the bot's box started the tick clear of blocks. This rule matches every warning and send-back in 40 recorded real-game runs (results/corner_scan.json). The corner check drops plans drifting over 0.25 blocks in the game's order, at no cost on validation. After both, the bots made 55/64 offline and in real Minecraft, the simulator's courses and ticks, with 0 corrections; 58 against 55 is within the search's seed noise.

7.9 Without the course list: greedy planner, learned router, learned route model

The policy is told the platform it stands on and the next three (section 5). Without that list, something else must fill these slots. The policy is not retrained, and only the goal's position is given (in the game, a beacon).

Greedy planner. At the start and after each landing on a new platform, parkour/planner.py (and bridge/planner.js, tested to plan identically) reads the solid blocks within 12 blocks. Platforms are standable one-block slabs; columns (wall tops, decoy

pillars) and head-hitter beams (a slab 4 blocks above a platform it touches) are not targets. Among unvisited platforms within the longest gap per height change that the feasibility study found possible (section 4) and with a clear straight flight path, it takes the one with the lowest air gap minus 2 cos(angle to the goal), and repeats from there for the later slots. Its failures are mostly routes: on 80 of its 92 failed standard test courses (80 of 114 with obstacles) the plan had left the course's order, usually by skipping ahead to a platform closer to the goal.

Learned router. The router learns this choice. At the same moments it picks the next platform among the greedy planner's own candidates, then the two after it; the unchanged final policy jumps. An attention network scores one token per platform in view (geometry, the greedy cost, flags). PPO trains it on these decisions as a semi-MDP (Sutton et al., 1999; discount 0.997 per tick): each decision earns the ticks' reward up to the next one, without the time cost and with -5 for a stall. An imitation term pulls it towards the course's next platform; the course is a training signal, never an input. A rule fixed before the chosen run (route-full-f) was trained froze the checkpoint with the best mean over the two 1,800-course validation sets among those that beat the planner on both (97.89% and 95.56%, equal to the course list; the best of 14 checkpoints). Played once on test, it beats the planner by 5.0 points over both sets (155 courses only it completes, 35 only the planner; McNemar p = 3e-19), 5.0 of the 5.4 points between the planner and the course list. Its wrong-platform failures fall from 80 and 80 to 12 and 8. Part of that gap is the test's fall rule, which follows the course's order: counting drops onto a later, lower platform as landings, the planner makes 94.7% and 92.4% and the router 98.0% and 95.2%. In real Minecraft, as with the offline bot, it completes exactly the simulator's courses with the same ticks and 0 server corrections; the ranking rests on the large simulator sets. Two controls on validation: PPO from scratch, without imitation, also beats the planner (+4.1 points) but declines after its best checkpoint; imitation alone matches PPO plus imitation (+0.2 points, p = 0.36), so PPO's share on top of imitation is not measurable.

Learned route model. noroute-v2 is a policy trained from scratch without the platform slots (3,000 iterations, obstacles on 30% of jumps, discount 0.99, chosen on validation).

It sees a wide coarse block scan and its last three landing spots; the platform list is used only in its training. It is below the greedy planner on both large sets. Obstacles fool it: 10 of its 23 real-game obstacle falls end standing on a beam or pillar.

Test results; the final policy jumps in the first three columns, and the real-game rows use the 60 + 60 courses of section 7.6:

Unseen test courses	Course list given	Greedy planner	Learned router	noroute-v2
1,200 standard, simulator	98.4%	92.3%	97.8%	85.4%
level 5 of those	93.5%	82.5%	92.0%	52.0%
1,200 with obstacles, simulator	95.2%	90.5%	95.0%	70.9%
60 standard, real Minecraft	59/60	51/60	57/60	55/60
60 with obstacles, real Minecraft	57/60	56/60	56/60	37/60

7.10 One network for every skill, and a gate

Besides the final policy (levels 0 to 5), each skill had a specialist: an ice fine-tune, the neo policy for levels 6 to 8 (section 7.7) and a fine-tune on the new jump kinds (head hitters, ice run-ups, block shapes, slime, ladders). One network (results/combined_policy.pt) trains on every skill with PPO plus a distillation term towards each course's specialist (the ice fine-tune for levels 0 to 5; Rusu et al., 2016). Its unweighted test mean over four groups (levels 0 to 5; the same levels with ice on 30% of platforms, family 16; levels 6 to 8; the new kinds; standard and obstacle sets pooled except on ice; simulator, played once) is 86.45%, against 83.70% for PPO alone on the same courses. It trails the final policy by about a point on levels 0 to 5 and on ice courses (95.8% and 96.8% against 96.8% and 97.8%) and leads the best specialist on levels 6 to 8 (66.8% against 63.9%) and the new kinds (86.4% against 85.3%; the final policy alone: 19.7%): a match, not a win. Offline and in real Minecraft it completes exactly the simulator's courses, with 0 server corrections: 59/60 and 59/60 of the final

policy's 120 test courses (section 7.6) and 59 of 65 fresh showcase courses of the new skills.

A gate (Jacobs et al., 1991), trained by PPO as a semi-MDP without imitation, instead picks one of seven frozen networks (the final policy, three specialists and three one-network candidates) at the start and at each landing. It beat the one network on validation, as its rule required (87.75% against 86.98%), and made 86.80% against 86.45% on test: a match, not a win. It runs in the simulator only.

8. Discussion and limitations

Why the transfer is exact. A Mineflayer bot computes its own movement with prismarine-physics, so a simulator that reproduces prismarine-physics reproduces the game as the bot experiences it. In 1,394 real-game episodes over 43 runs (results/games/) and 120 router episodes (results/routing/), every outcome but one equals the simulator's. That one and the only 2 server corrections came from one run before the corner check, next to neo-wall corners, where the game orders collisions differently from prismarine-physics (section 7.8). The claim is about a Mineflayer bot on a local server: we matched prismarine-physics, not the game's code, and tested neither network latency nor a remote server.

Generalisation. Disjoint seed families make "unseen" measurable. The final policy completes 98.4% of the standard test courses and 93.0 to 98.3% in course styles outside its training (simulator); level 5 holds most of its failures. Being told the platform chain matters (82.4% without it, 93.3% with, section 7.2), but the chain need not come from a course list: a router that imitates the list in training but sees only nearby blocks and the goal at play closes 5.0 of the 5.4 points between the greedy planner and the course list on test (simulator), and most of its remaining failures are failed jumps. A policy that learns route and jumps together stays below the greedy planner on both test sets (section 7.9).

Reward terms interact with risk. The fall penalty, our height term, helped most on level 5 (72.0% with it, 55.2% without, section 7.2), but on jumps that rarely land it taught the agent to refuse: a stall cost almost nothing, a failed jump the full penalty (section

7.7). Whether a stall counts as a failure is part of the reward design, not a detail of the environment: a penalty that helps on the jumps an agent can make also decides which jumps it tries.

Search against learning. PPO never found a neo by exploring. Imitating a search expert, then DAgger, taught the jump, but on the single-neo test courses the best policy alone completes 42.8%, while the neo planner, the same kind of search run at each takeoff during play, completes 86.25% (strict, simulator, played once; section 7.8). Both need a simulator exact and cheap enough to search from any state. The price is search time at every takeoff (2.8 s per search on the CPU in the test), and a skill that is planned rather than learned.

Statistics. Many results rest on one training run: most hyperparameter settings, each router run, the one network and the gate. The neo planner's search seed adds noise larger than the single-neo test's margin (section 7.8). Validation peaks are optimistic (the router is the best of 14 checkpoints, the gate the best of 15 validations); the tests, played once, are the unbiased numbers. The 60-course bot runs check that the game reproduces the simulator; they cannot rank policies a few points apart.

Limitations.

- The final policy trained on stone only; the other blocks are opt-in (section 3), and water is not modelled.
- The greedy planner and the learned router were built and tested on courses without neos.
- The feasibility study uses scripted strategies from rest, so it underestimates what is possible (section 7.4) and what chained jumps allow.
- Courses are built and reset with server commands (section 6); a carpet over air breaks in the game, so the bot puts a structure void under it.
- With the neo planner the bot pauses while it searches, so those episodes are not in real time.
- Runs used an RTX 5080 and, for the learned router, the teammate's RTX 4060 Ti.

9. Conclusion

A PPO agent trained on generated courses, in a simulator that reproduces a Mineflayer bot's movement engine, completes parkour courses it has never seen: 98.4% of 1,200 held-out standard courses and 95.2% of 1,200 obstacle courses (simulator; level 5: 93.5% and 81.0%). In real Minecraft the bot made exactly the simulator's courses (59 of 60 standard, 57 of 60 obstacle), with 0 server corrections and replays within 2.1e-14 blocks of the game. Away from wall corners, where the game orders collisions differently, the gap in movement between simulation and the game is rounding error.

Of our first reward ideas, the fall penalty mattered most (level 5: 72.0% against 55.2%) and distance shaping made learning about three times faster; neither the landing bonus nor the curriculum changed the final score reliably. On neo jumps, which random tries rarely land, the same penalty made refusing cheaper than trying: PPO alone never made one. Search demonstrations and DAgger raised the completed level 6-8 neo test courses from 0 to 29.0% (simulator). Searching each takeoff at play time with the exact simulator met the 85% single-neo goal, narrowly: 276 of 320 test courses (86.25%, strict, simulator, played once) against 42.8% for the best policy alone; in real Minecraft the bot made the simulator's courses on the first 64 of them (58, then 55 after two later fixes; section 7.8), with 0 corrections. No double or triple neo landed in 64 searched layouts: evidence, not proof, that they are impossible.

Without the course list, a learned router trained with PPO plus an imitation term towards the course's next platform picks each platform from the blocks the bot sees, and the unchanged final policy jumps: 97.8% and 95.0% of the standard and obstacle test courses (simulator, played once), against 92.3% and 90.5% for the greedy planner, and 57 and 56 of 60 in real Minecraft.

One network trained with PPO and distillation plays every skill group about as well as that group's best specialist (test mean 86.45%, simulator): a match, not a win. It completes 86.4% of the test courses with new jump kinds and blocks (the final policy alone: 19.7%).

Next steps are neo courses without the course list, more blocks, water first, and real hand-built maps.

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